Fantasy Premier League Points Forecasting and Squad Optimization

Project status report

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Abstract. This report documents a Fantasy Premier League (FPL) points-prediction and squad-optimization system: a per-position ensemble of machine-learning regressors feeding a mixed-integer linear program (MILP) that selects squads, transfers, captaincy and chips. Originally a Master's thesis (LSTM in R + MILP in Python), it has been rewritten as a single weekly-runnable Python pipeline and validated against the original system on an identical backtest: 1,966 realized points vs. the LSTM's 1,526 (+29%) over the same 31 gameweeks. All experiments - including the ones that failed, and one unresolved regression - are reported.

1 Introduction

The system predicts FPL player points one to several gameweeks ahead and converts those forecasts into weekly squad decisions via a MILP based on Kristiansen et al. (2018). The guiding principle of the current phase is honest empirical validation: every candidate technique is tested against the production system on a fixed backtest, and negative results are reported rather than discarded. This report gives the data and methodology, the accumulated results, and the known methodological limitations identified by an internal code and concept review.

2 Data

Per-gameweek player statistics are sourced from Vaastav Anand's public mirror of the official FPL API (`vaastav/Fantasy-Premier-League` on GitHub), covering six seasons - 2020-21 through 2025-26, 162,981 player-gameweek rows. The two oldest seasons lack the Opta expected-goals family and the `starts` column (introduced 2022-23); these are treated as missing, which the tree-based models handle natively. Gameweeks are indexed by a single ascending counter (`GW_global`): season N in the dataset occupies gameweeks $(N-1) \times 38 + 1$ through $N \times 38$, so the 2024-25 season's gameweeks 1-31 - the validation window used throughout - is GW153-183.

Exploratory analysis (`notebooks/eda.ipynb`, all six seasons) establishes the two facts the methodology is built around. First, `total_points` is heavily right-skewed and zero-inflated at every position (Shapiro-Wilk rejects normality at $p < 0.001$ throughout), which motivates a scale-free error metric (Section 3.4) and models that do not assume Gaussian errors. Second, the distribution is stable across seasons (per-position means for players who featured vary within roughly 2.4-3.8 points across all six seasons, medians flat at 1-2), with one caveat: over six seasons the average defender score-per-gameweek series no longer passes an Augmented Dickey-Fuller stationarity test ($p = 0.41$; GK/MID/FWD all reject the unit root at 5%), plausibly reflecting the 2025-26 `defensive_contribution` scoring change - a reason for caution with fixed-mean time-series baselines at DEF specifically.

3 Methodology

3.1 Feature engineering

Each player-gameweek row carries roughly 80 engineered features:

- **Form features.** Rolling 3- and 5-gameweek means, previous-gameweek value, and a season-to-date expanding mean over 18 per-gameweek statistics (points, minutes, xG family, BPS, ICT index, price, ownership, etc.). All are shifted one gameweek relative to the target row, so no row's features contain its own outcome.
- **Fixture-difficulty features.** The official FPL Fixture Difficulty Rating (FDR, 1-5) of the opponent faced that gameweek, and the mean FDR over this plus the next two scheduled fixtures. These are deliberately *not* shifted: fixture lists are published before gameweeks are played, so they are known-ahead inputs, not leakage.
- **Minutes-projection features.** Rolling 5-game start rate and 60+-minute-appearance rate, shifted like the form features - a "will he actually play?" signal, since a benched player scores approximately zero regardless of ability.

3.2 Forecasting models

Four independent models are trained, one per position (GK/DEF/MID/FWD): the statistics that predict a goalkeeper's points are largely disjoint from a forward's, and pooling risks one position's scale dominating the loss.

Per position, every regressor in a fixed registry is trained on the same features: LightGBM, XGBoost, CatBoost, plain OLS, Ridge, ElasticNet, partial least squares (PLS), Random Forest, Extra Trees, k-nearest neighbors, linear support-vector regression (LinearSVR), and a sample-capped RBF-kernel SVR. Gradient-boosted models receive raw features including missing values; linear, kernel and distance models are wrapped in a zero-imputer and standardizer. Registry members that preliminary tests found unhelpful are retained rather than deleted - the blending step (below) simply assigns them near-zero weight, and keeping them preserves the comparison for later revisiting.

The registry's predictions are combined by non-negative least squares (NNLS): weights $w \geq 0$ minimizing $\left\|\sum_i w_i \hat{y}_i - y\right\|$, normalized to sum to one. Two design choices matter methodologically: (a) weights are fit on one half of the held-out test window and the ensemble is evaluated on the *other* half, so reported ensemble accuracy is a genuine holdout figure; (b) plain OLS is designated the *index* - the simple benchmark every technique must beat, in the sense a passive market index is the bar an active strategy has to clear.

Alongside the point-forecast ensemble, per-position LightGBM quantile regressors (p10/p50/p90) produce a prediction interval per player-gameweek. Quantile crossing is repaired by row-wise sorting. This probabilistic view exists because two players with equal expected points are not equal decisions - upside matters on the 2x captain multiplier - but it does not yet feed the optimizer.

3.3 Evaluation design

Three complementary layers, in increasing order of decision-relevance:

1. **Static split.** Train on GW $\leq$ 152, evaluate on GW153-183 - the same 2024-25 GW1-31 window the original LSTM was validated on, keeping every comparison in this report on one fixed window.
2. **Walk-forward validation.** For each gameweek from a start point, train on strictly earlier data and predict that gameweek only, rolling forward - many genuinely out-of-sample evaluations rather than one.
3. **Actual-points backtest (gold standard).** Walk-forward predictions are fed through the MILP over GW153-183, and the resulting squads are scored with *realized* points. This is the check that matters: an accuracy gain that does not survive contact with the optimizer is not an improvement (Section 4.3 shows exactly this happening).

3.4 Error metrics

Point forecasts are scored with MAE and, primarily, MASE (Hyndman & Koehler, 2006): MAE divided by the in-sample MAE of a naive "same as last gameweek" forecast, with the scale fit on training data only. For a zero-inflated target, raw MAE is uninterpretable in isolation; MASE $<$ 1 means the model beats the naive floor regardless of the target's scale. Probabilistic forecasts are scored with pinball loss (the proper scoring rule for quantiles) and the empirical coverage of the [p10, p90] band against its nominal 0.80.

3.5 Squad optimization

The MILP follows Kristiansen et al. (2018): a 15-player squad (2 GK / 5 DEF / 5 MID / 3 FWD) under a budget, at most 3 players per club, an 11-player lineup under formation constraints, captain and vice-captain, a limited number of free transfers with a 4-point penalty per extra transfer, and one-shot chip logic (two wildcards, free hit, bench boost, triple captain). It is solved as a rolling horizon: re-solved each gameweek over a configurable lookahead, with only the first gameweek's decision locked in. Venter & van Vuuren (2024) - whose formulation matches this one almost exactly - attribute their strong case-study result (top 4.08% of 8.24M managers) to lookahead and forecast quality rather than optimizer sophistication, which is why this project's effort concentrates on forecasting.

4 Results

4.1 Backtest lineage

All systems scored by realized points over the same 2024-25 GW1-31 window, identical optimizer:

System	**Actual points**
Old LSTM + MILP (thesis)	1526
LightGBM + MILP, 4-season history	1811
Ensemble + MILP, 4-season history	1900
Ensemble + MILP, 6-season history	**1966**
1. fixture/minutes features	1880[1]

The best validated configuration (1,966) improves on the thesis system by 29%. Extending history from four to six seasons was an unambiguous win: MASE fell at every position (FWD below 1.0 for the first time in the project), and realized points rose from 1,900 to 1,966 (+3.5%).

[1]Unresolved regression despite improved forecast accuracy - see Section 4.3.

4.2 Forecasting-technique experiments

Simple and econometric baselines - rolling mean, naive drift, SES, Holt, Theta, Croston, pooled AR(1), per-player ARIMA(1,0,1) - were each tested per position against the ensemble. None beats it anywhere; SES is the best of the simple methods, Croston the worst (its intermittent-demand design reacts too slowly when a player goes cold, and pooling it by position hides the per-player heterogeneity the source literature exploited). Full tables in `RESEARCH_LOG.md`. Plain OLS - the designated index - beats every one of these simple baselines, and the ensemble in turn beats OLS at every position on four-season data; on six-season data OLS narrowly overtakes the ensemble at GK only (MASE 0.579 vs 0.595), plausibly because goalkeeping is a small, simple-signal position that gains more from data volume than from ensemble complexity.

A preliminary standalone check of the newest registry members (static split, current feature set) found **LinearSVR beating the full ensemble at every position** (e.g. DEF 0.713 vs 0.799, FWD 0.869 vs 0.959 MASE) - the largest single-model result so far, plausibly because its epsilon-insensitive loss tracks the MAE-family metrics better than squared-error losses on a zero-inflated target with outlier hauls. XGBoost and RBF SVR underperformed and are retained as near-zero-weight blend candidates. Full-ensemble re-evaluation with the expanded registry was in progress at the time of writing.

4.3 Fixture/minutes features: better forecasts, worse squads

Adding the fixture-difficulty and minutes-projection features improved MASE at every position (GK 0.595→0.588, DEF 0.830→0.799, MID 0.811→0.799, FWD 0.984→0.959; DEF most, consistent with clean-sheet dependence) - yet the actual-points backtest *fell* from 1,966 to 1,880. The leading hypothesis: the features smooth the mean forecast toward safe, nailed players, and the MILP - which maximizes expected points and is blind to variance - loses the high-ceiling captaincy differentials that drive realized hauls. The features are committed but explicitly not declared a net win; resolving this divergence (noise check on a second window, or feeding the probabilistic upside signal into captaincy) is the top open item. This is the project's clearest demonstration of why the actual-points backtest, not MASE, is the gold-standard check.

The probabilistic module's [p10, p90] coverage is 0.88-0.93 against the nominal 0.80 - somewhat over-wide intervals, expected where the p10 quantile pins at zero for blank-prone players. Usable for relative risk ranking; conformal calibration is a possible refinement.

4.4 Exploratory branches

Two isolated branches, neither merged: a Bayesian belief-state MDP manager after Matthews et al. (2012) (1,083 points myopic / 847 Q-learning on the four-season window where the production system scored 1,900 - expected, given necessary simplifications documented on the branch), and a per-player model-selection plan (Venter & van Vuuren's per-player method choice; feasibility capped by 35% of live players having no prior-season history - recommendation: single-position pilot before any commitment).

5 Known methodological limitations

An internal review of the code and modeling concepts (2026-07-04) identified the following. None invalidates the relative comparisons above; two mildly inflate absolute backtest numbers, and two affect live use only. All are tracked in `TODO.md`.

1. **Blend-weight leakage into the backtest window.** Ensemble blend weights are fit on the first half of GW153-183 and then reused by the walk-forward backtest over that same window, so the first half's predictions use weights fit on their own outcomes. Absolute

headline numbers (1,966 / 1,880) are modestly optimistic; the comparison between them is fair since both leak identically. Fix: fit backtest weights strictly before the window.

2. **Index-check asymmetry.** The ensemble's MASE is (correctly) measured on the test window's second half only, but the OLS index's on the full window - the "beats index" verdict compares slightly different evaluation sets.
3. **Live-mode feature staleness.** The weekly driver re-uses each player's last played row for future gameweeks, so (a) live fixture-difficulty features describe last week's fixture rather than the upcoming one, and (b) rolling form excludes the player's most recent match. Backtests are unaffected; live recommendations are degraded until fixed.
4. **Optimizer rule currency.** The MILP caps banked free transfers at 2 (the pre-2024-25 rule; now 5) and assumes players sell at current market value (real FPL: purchase price plus half the profit). Fine for historical comparison, wrong for live 2026-27 play.

6 Conclusion and future work

The rewritten pipeline beats its thesis-era predecessor by 29% in realized backtest points, and the phase's central negative finding is itself useful: no simpler technique tested - econometric or ML - beats the per-position ensemble, so further gains must come from architecture (per-player selection, probabilistic captaincy) rather than from adding model types. Priorities, in order: resolve the fixture/minutes MASE-vs-points divergence; fix the blend-weight leakage and live-mode staleness from Section 5; evaluate LinearSVR inside the full ensemble; then the per-player selection pilot.

7 References

Croston, J. D. (1972). Forecasting and stock control for intermittent demands. *Journal of the Operational Research Society*, 23(3), 289-303.

Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4), 679-688.

Kristiansen, B. K., Gupta, A., & Eilertsen, W. (2018). *Developing a forecast-based optimisation model for Fantasy Premier League* (MSc thesis). Norwegian University of Science and Technology, Trondheim.

Matthews, T., Ramchurn, S. D., & Chalkiadakis, G. (2012). Competing with humans at fantasy football: Team formation in large partially-observable domains. *Proceedings of the Twenty-Sixth AAAI Conference on Artificial Intelligence*, 1394-1400.

Venter, V., & van Vuuren, J. H. (2024). An optimisation approach towards soccer Fantasy Premiere League team selection. *ORiON*, 40(1), 69-107.